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BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Ethical Analysis Report

Group ID: 25-26J-091

Project Title: Smart Real-Time Age, Gender, Emotion, and Behavior-Based Advertising System for Sri Lankan Malls

	Identification and Analysis
Data Privacy Concerns	System captures facial data to infer age, gender, and emotional states, which raises significant privacy concerns. To address this, the system is designed to avoid storing personally identifiable information (PII) such as raw facial images whenever possible. Instead, temporary processing is performed in real-time, and only aggregated or anonymized insights are stored. Additionally, since the system operates in a public mall environment, clear signage and user awareness mechanisms are necessary to inform individuals that analytics are being used. Where applicable (e.g., interactive kiosk features), explicit user consent is obtained before capturing or processing data.
Data Security	System incorporates multiple layers of security to protect collected data. All communication between system components (camera, backend, database) is secured using encrypted protocols (HTTPS/SSL). Sensitive data such as embeddings or analytics logs are stored in secured databases (e.g., MongoDB with access control and authentication). Access to the system is restricted through role-based authentication (Admin vs Kiosk users), ensuring that only authorized personnel can view or manage analytics data. Additionally, regular system monitoring and logging mechanisms help detect and prevent unauthorized access or misuse.
Bias and Fairness	The AI models used in System (age, gender, emotion,engagement) may exhibit biases due to imbalanced training datasets, especially with respect to ethnicity, lighting conditions, or cultural expressions. To mitigate this: •The system is tested using diverse datasets, including representations closer to the Sri Lankan demographic. •Techniques such as weighted region learning and model fine-tuning are used to improve fairness. •Outputs are treated as probabilistic estimates rather than absolute truths, reducing the risk of misclassification impact.

	Continuous evaluation is necessary to ensure that the system does not unfairly favor or misrepresent specific groups.
Transparency and Accountability	System maintains transparency by clearly documenting: •The data flow pipeline (camera → detection → inference → recommendation) •The models used (CNN-based age/gender/emotion/engagement models, recommendation engine) •The decision logic behind advertisement selection The system does not make critical decisions affecting individuals directly; instead, it provides aggregate-level insights for advertisers. However, the development team remains accountable for: •Ensuring ethical deployment •Monitoring system behavior •Addressing any unintended consequences Clear documentation and reporting ensure that stakeholders understand how the system operates.
Impact on Stakeholders	Customers (Mall Visitors) •Positive: More relevant and engaging advertisements •Concern: Potential discomfort due to being analyzed without direct interaction Advertisers •Positive: Improved targeting and return on investment (ROI) •Risk: Over-reliance on AI predictions that may not always be accurate Mall Management •Positive: Enhanced customer experience and smarter space utilization •Risk: Responsibility for ensuring ethical deployment and compliance Developers / Researchers •Responsible for maintaining ethical AI practices, reducing bias, and ensuring compliance with privacy regulations. Overall, while system provides strong commercial and analytical benefits, it must be deployed with careful consideration of privacy, fairness, and transparency to avoid negative societal impacts.